Laboratory 1 Report

Environment Setup: Anaconda, Jupyter Notebook and Git

Field	Detail
Course	Introduction to Artificial Intelligence (AI-101L)
Instructor	Atteq ur Rehman
Student	Hadia Imran
GitHub	hadia-web11
Repository	https://github.com/hadia-web11/ai-lab-portfolio
Lab date	23 October 2026 (notebook); Git work completed 13 November 2026

1. Installation evidence

Miniconda3 (installer py314_26.9.4-0, 64-bit) was installed on Windows. A dedicated environment **ai-lab** (Python 3.11) was created for the course and registered as a Jupyter kernel, as specified in Task 1.1. Git 2.56.0.4 for Windows was installed with VS Code as the editor and **main** as the default branch name.

```
conda create -n ai-lab python=3.11 -y
conda activate ai-lab
pip install jupyter ipykernel pandas
python -m ipykernel install --user --name ai-lab --display-name "Python (ai-lab)"

git config --global user.name  "Hadia Imran"
git config --global init.defaultBranch main
```

Notebook output: Python prints the greeting; pandas version reported as **3.1.1**.

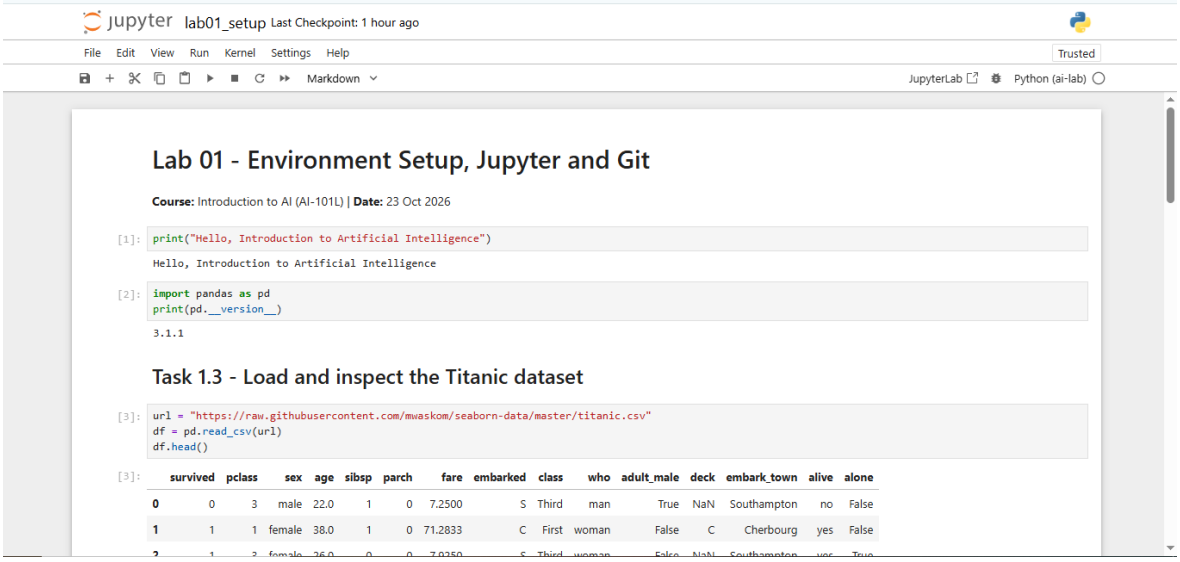


Figure 1. lab01_setup.ipynb running on the Python (ai-lab) kernel (top right): greeting cell, pandas 3.1.1 and the Titanic head() output.

Column	Missing values
deck	688
age	177
embarked	2
embark_town	2

The seaborn version of the file has four incomplete columns (embark_town duplicates embarked), so the "three columns" in the exercise are deck, age and embarked, with embark_town as the repeat of embarked.

In-lab exercise 3: largest standard deviation to mean ratio

`sibsp` and `parch` have the largest ratio (about 2.1), followed by `fare` (about 1.5). A column whose spread is large relative to its mean is dominated by a few extreme values. Without scaling, distance-based and gradient-based models give such columns disproportionate influence, so features need standardising first.

Full inspection reports for Titanic and Penguins are in lab01/inspect_reports.txt.

3. Git and GitHub work

The repository **ai-lab-portfolio** holds each lab in its own folder. The first commit published `lab01/` and `.gitignore`. Three commits were then made on branch `lab01/exercises` and merged into main (in-lab exercise 4). Because main had not moved, Git performed a fast-forward merge, so the graph is a straight line.

Output of git log --oneline --graph --all

```
* 62c65b0 (HEAD -> main) Add exercise 3 note
* be245f3 Add exercise 2 note
* b6d6336 Add exercise 1 note
* bd103d0 (origin/main) Lab 01: environment setup and first notebook
```

Push to GitHub: `bd103d0..62c65b0 main -> main`. Report update pushed afterwards as commit 8c1d6f1.

Commit history

Commit	Message / purpose
bd103d0	Lab 01: environment setup and first notebook
b6d6336	Add exercise 1 note (branch lab01/exercises)
be245f3	Add exercise 2 note
62c65b0	Add exercise 3 note (fast-forward merged into main)
8c1d6f1	Add git graph to report
5a3b233	Add requirements.txt (branch lab01/homework)
7fe0817	Merge commit of pull request #1 into main

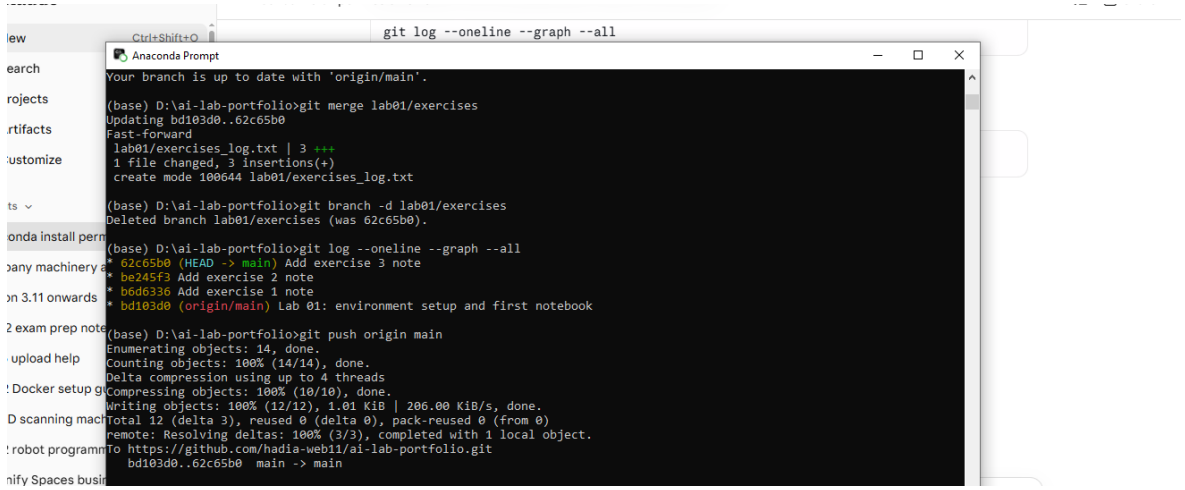


Figure 2. Terminal: merge of lab01/exercises, git log --oneline --graph --all, and push to origin/main.

4. Home assignment and links

`lab01/inspect.py` defines `inspect(path_or_url)`, which reports shape, dtypes, missing counts and percentages, and the numeric summary. It was run on two public datasets (Titanic and Penguins); the results are in `lab01/inspect_reports.txt`. The work was merged through a GitHub pull request from branch `lab01/homework`.

Item	URL
Repository	https://github.com/hadia-web11/ai-lab-portfolio

Item	URL
Pull request #1 (merged)	https://github.com/hadia-web11/ai-lab-portfolio/pull/1
Merge commit	commit 7fe0817 on main (open from the pull request page)
Report file	https://github.com/hadia-web11/ai-lab-portfolio/blob/main/lab01/report.md

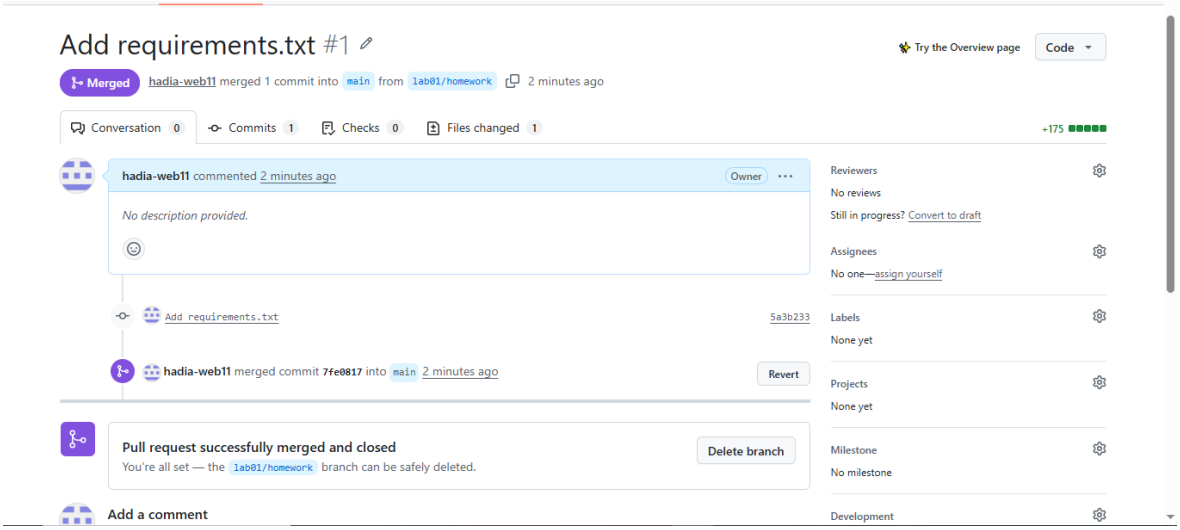


Figure 3. Pull request #1 merged into main from lab01/homework (merge commit 7fe0817).

5. Deliverables checklist

Deliverable	Status
lab01/lab01_setup.ipynb	Committed
lab01/inspect.py	Committed
lab01/report.md	Committed
.gitignore	Committed
requirements.txt	Committed in PR #1
Notebook exported to .html and .pdf	Appendix A (notebook export, pages 5 onward)
Git graph in report	Included (section 3)

Appendix A: Notebook export (lab01_setup.ipynb)

Rendered from the exported HTML of the notebook, run on the Python (ai-lab) kernel. Cells appear in execution order.

Lab 01 - Environment Setup, Jupyter and Git

Course: Introduction to AI (AI-101L) | Date: 23 Oct 2026

Input

```
print("Hello, Introduction to Artificial Intelligence")
```

Output

```
Hello, Introduction to Artificial Intelligence
```

Input

```
import pandas as pd
print(pd.__version__)
```

Output

```
3.1.1
```

Task 1.3 - Load and inspect the Titanic dataset

Input

```
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/titanic.csv"
df = pd.read_csv(url)
df.head()
```

Output

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.25	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.925	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.05	S	Third	man	True	NaN	Southampton	no	True

Input

```
df.info()
```

Output

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    str
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    str
 8   class        891 non-null    str
 9   who          891 non-null    str
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    str
 12  embark_town  889 non-null    str
 13  alive        891 non-null    str
 14  alone        891 non-null    bool
dtypes: bool(2), float64(2), int64(4), str(7)
memory usage: 92.4 KB
```

Input

```
df.describe()
```

Output

	survived	pclass	age	sibsp	parch	fare
count	891.0	891.0	714.0	891.0	891.0	891.0
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.0	1.0	0.42	0.0	0.0	0.0
25%	0.0	2.0	20.125	0.0	0.0	7.9104
50%	0.0	3.0	28.0	0.0	0.0	14.4542
75%	1.0	3.0	38.0	1.0	0.0	31.0
max	1.0	3.0	80.0	8.0	6.0	512.3292

Input

```
print(df.shape)
```

Output

```
(891, 15)
```

In-Lab Exercise 2 - columns with missing values (from info())

Input

```
missing = df.isnull().sum()
missing[missing > 0].sort_values(ascending=False)
```

Output

```
deck           688
age            177
embarked         2
embark_town      2
dtype: int64
```

Three columns contain missing values: deck (688), age (177) and embarked / embark_town (2 each; embark_town is the text version of embarked).

In-Lab Exercise 3 - largest std/mean ratio (from describe())

Input

```
d = df.describe().T
d["std_over_mean"] = d["std"] / d["mean"]
d[["mean","std","std_over_mean"]].sort_values("std_over_mean", ascending=False)
```

Output

	mean	std	std_over_mean
parch	0.381594	0.806057	2.112344
sibsp	0.523008	1.102743	2.108464
fare	32.204208	49.693429	1.543073
survived	0.383838	0.486592	1.267701
age	29.699118	14.526497	0.489122
pclass	2.308642	0.836071	0.362149

Answer: sibsp and parch have the largest ratio (about 2.1), then fare (about 1.5). A large ratio means the values are very spread out relative to their average, and columns such as fare (0 to 512) are on a much bigger scale than age or pclass. A model that is not given scaled inputs (e.g. gradient descent, KNN, SVM)

will be dominated by the large-scale columns, so training is slow or biased.

Task 1.4 - Export

File -> Save and Export Notebook As -> HTML / PDF.